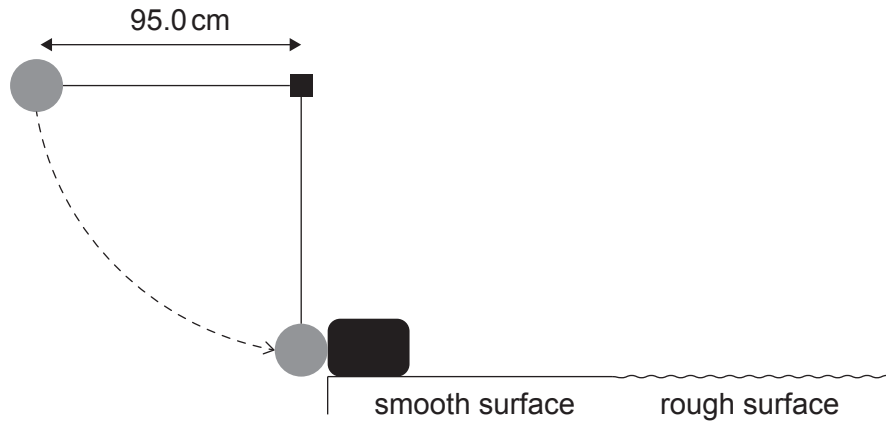


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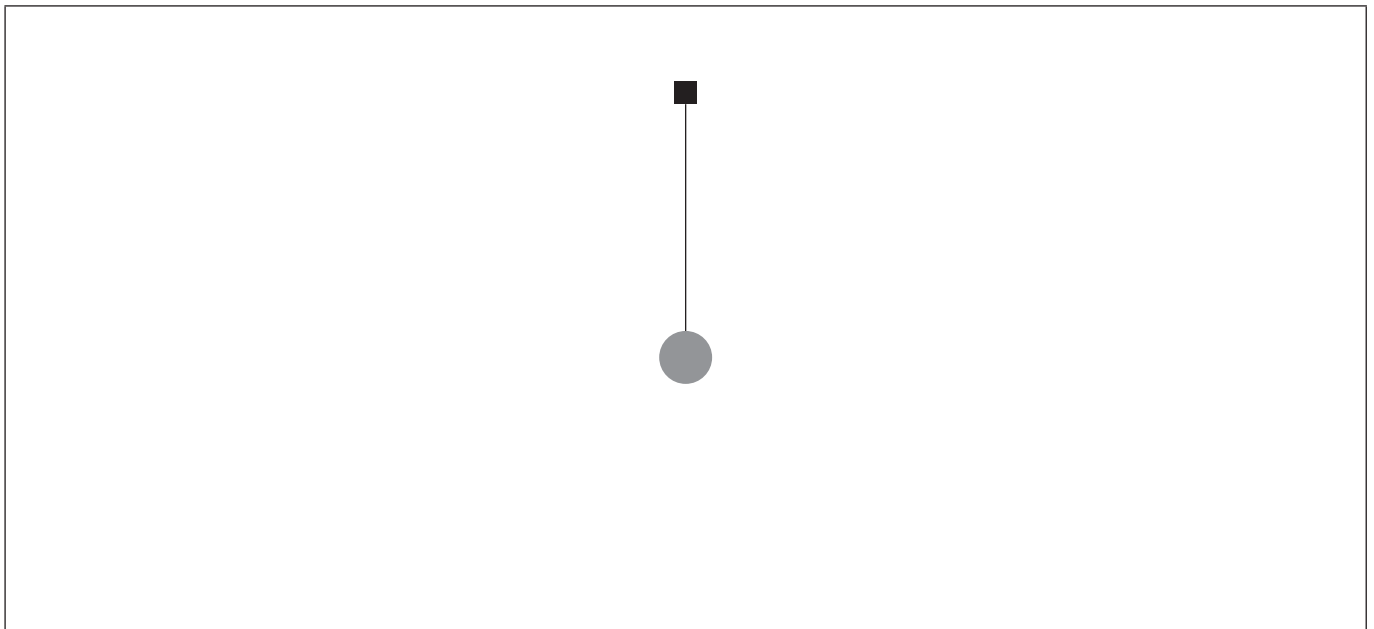
9. A ball of mass 0.800 kg is attached to a string. The distance to the centre of the mass of the ball from the point of support is 95.0 cm . The ball is released from rest when the string is horizontal. When the string becomes vertical the ball collides with a block of mass 2.40 kg that is at rest on a horizontal surface.



- (a) Just before the collision of the ball with the block,

- (i) draw a free-body diagram for the ball.

[2]



(This question continues on the following page)

PAPER 2 / QUESTION 9

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(Question 9 continued)

- (ii) show that the speed of the ball is about 4.3 m s^{-1} . [1]

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- (iii) determine the tension in the string. [2]

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(This question continues on the following page)

(Question 9 continued)

(b) After the collision, the ball rebounds and the block moves with speed 2.16 m s^{-1} .

(i) Show that the collision is elastic.

[4]

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(ii) Calculate the maximum height risen by the centre of the ball.

[2]

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(c) The coefficient of dynamic friction between the block and the rough surface is 0.400.

Estimate the distance travelled by the block on the rough surface until it stops.

[3]

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